BloodLink: A Technology-Enabled Blood Donor Matching and Notification System for Improved Blood Donor Accessibility

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I. Introduction

Blood is an essential resource in healthcare because it is needed for patients experiencing severe bleeding, trauma, surgery, childbirth complications, severe anemia, and other medical conditions. Despite the importance of blood transfusion, many patients still do not have timely access to safe blood. The World Health Organization (WHO) reported that approximately 120.4 million blood donations were collected globally, with significant differences in donation rates between high- and low-income countries. WHO also emphasizes that a reliable blood supply depends on a stable base of regular, voluntary, unpaid blood donors.

In the Philippines, Republic Act No. 7719, or the National Blood Services Act of 1994, promotes voluntary blood donation and aims to provide an adequate, safe, affordable, and equitable supply of blood and blood products. The law also recognizes the importance of organized blood-service networks and voluntary donor programs.

However, one challenge that may affect timely donor coordination is the difficulty of connecting people who need blood with potential donors. Blood requests are often communicated through personal contacts, social media, or separate communication channels. This creates an opportunity for technology to improve how verified blood requests and potential donors are connected.

Therefore, this project proposes BloodLink, a blood donor matching and notification platform designed to support the communication and coordination between authorized blood-service facilities and registered potential donors.

II. Problem Description

The identified problem is the lack of accessible and timely communication between blood requests and potential donors. During emergencies or planned procedures, families may depend on personal networks or social-media posts to locate donors. This can make the search fragmented, time-consuming, and difficult to organize.

Key challenges include:

• Limited visibility of urgent blood requests to potential donors.
• Difficulty identifying potentially suitable donors based on blood type and location.
• Delays in communicating verified requests.
• Donor information being scattered across different communication channels.
• Need for safer coordination with authorized blood-service facilities.

WHO's 2026 data show that 55 countries reported fewer than 10 blood donations per 1,000 population, illustrating continuing differences in blood availability. WHO also reports that more than 85% of estimated global blood donations in 2023 came from voluntary unpaid donors.

III. Project Solution

The proposed solution is the development of
BloodLink: a Blood Donor Matching and Notification

Platform

The system is designed to help authorized hospitals, blood banks, or blood-service facilities publish verified blood requests and notify registered potential donors who may be suitable based on basic matching information such as blood type, location, and stated availability.

BloodLink is intended to function as a coordination and communication tool. It does not replace professional donor screening, blood typing, compatibility testing, infectious-disease screening, collection, storage, or transfusion decisions. These procedures remain under qualified and authorized blood-service professionals.

How BloodLink Works

1. REGISTER
Donor creates a profile with blood type, location, availability, and relevant donation history.

2. VERIFY REQUEST
An authorized facility posts or validates a blood request, including blood type, units needed, location, and urgency.

3. MATCH
The system identifies potentially suitable registered donors using selected matching criteria.

4. NOTIFY
Potentially suitable nearby donors receive an alert and can respond through the platform.

5. COORDINATE & DONATE
The donor follows instructions from the authorized blood-service facility

IV. Conclusion

The lack of timely access to potential blood donors is a communication and coordination challenge that can affect the process of obtaining blood when it is needed. Although existing blood-service organizations and voluntary donation programs play an essential role, technology can provide an additional method for improving communication between verified blood requests and potential donors.

BloodLink proposes a practical technology-enabled solution through a centralized platform for donor registration, verified blood requests, potential donor matching, and notifications. By making donor communication more organized and accessible, the proposed system can support voluntary blood donation and improve coordination with authorized blood-service facilities.

Ultimately, BloodLink is intended to support existing blood-service systems rather than replace them. Its goal is to use technology to help connect potential donors with verified needs while maintaining the professional standards and safety procedures required for blood donation and transfusion.

Reference:

World Health Organization. (2026, June 12). *Blood safety and availability.* WHO – Blood safety and availability

World Health Organization. (2026, June 12). *Safe blood supply improves as voluntary donations exceed 85%, but many people still lack access.* WHO – World Blood Donor Day 2026

Republic of the Philippines. (1994). *Republic Act No. 7719: National Blood Services Act of 1994.* Republic Act No. 7719 – Lawphil

World Health Organization Regional Office for South-East Asia. (2023). *Voluntary non-remunerated blood donations to ensure blood safety in the WHO South-East Asia Region to support universal health coverage.* WHO South-East Asia – Voluntary Blood Donations

Department of Health. (2005). *Administrative Order No. 2005-0002: Rules and Regulations for the Establishment of the Philippine National Blood Services.* DOH Administrative Order No. 2005-0002